

MATH 39 — Linear Algebra

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1. Systems of Linear Equations and Matrices

1.1 Linear Equations

Definition A **linear equation** in the variables $x_1, x_2, \dots, x_n$ is an equation that can be written in the form

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b$$

where $a_1, a_2, \dots, a_n$ and b are constants. The constants $a_1, a_2, \dots, a_n$ are called the **coefficients** of the equation, and b is called the **constant term**.

Examples of linear equations include:

- $2x + 3y - z = 5$
- $-x_1 + 4x_2 + 0x_3 = 7$
- $0x + 0y + 0z = 0$

Non-examples of linear equations include:

- $x^2 + y = 3$ (because of the x^2 term)
- $xy + z = 4$ (because of the xy term)
- $\sin(x) + y = 1$ (because of the $\sin(x)$ term)
- $x + \sqrt{y} = 2$ (because of the $\sqrt{y}$ term)

Systems of Linear Equations

Definition A **finite** collection of linear equations involving the same set of variables is called a **system of linear equations**.

Example of a system of linear equations:

$$\begin{cases} 2x + 3y - z = 5 \\ -x + 4y + 2z = 7 \\ 3x - y + z = 2 \\ \dots \end{cases}$$

Definition A sequence of numbers $(s_1, s_2, \dots, s_n)$ is called a **solution** to the system of linear equations if, when we substitute s_i for x_i in each equation, the equation holds true.

Example Consider the system of equations:

$$\begin{cases} x + y + z = 3 \\ 2x + y + 3z = 1 \end{cases}$$

Is the sequence $(-2, -5, 0)$ a solution to this system? Does $x = y = z = 1$?

Solution To check if $(-2, -5, 0)$ is a solution, substitute $x = -2$, $y = -5$, and $z = 0$ into both equations:

- First equation: $(-2) + (-5) + 0 = -7 \neq 3$ (does not hold)
- Second equation: $2(-2) + (-5) + 3(0) = -4 - 5 = -9 \neq 1$ (does not hold)

Thus, $(-2, -5, 0)$ is not a solution to the system.

Next, check if $(1, 1, 1)$ is a solution:

- First equation: $1 + 1 + 1 = 3$ (holds)
- Second equation: $2(1) + 1 + 3(1) = 2 + 1 + 3 = 6 \neq 1$ (does not hold)

Thus, $(1, 1, 1)$ is also not a solution to the system.

Solutions to Linear Systems

Definition There are three possible scenarios for the solutions to a system of linear equations:

- **No solution:** The system is inconsistent, meaning there are no values for the variables that satisfy all equations simultaneously.
- **Unique solution:** There is exactly one set of values for the variables that satisfies all equations.
- **Infinitely many solutions:** There are multiple sets of values for the variables that satisfy all equations. (i.e. the same line or scalar multiples of each other)

Systems of linear equations that have at least one solution are called **consistent**, while those with no solutions are called **inconsistent**.

Matrices

Definition A **matrix** is a rectangular array of numbers arranged in rows and columns. The numbers in the matrix are called its **entries** or **elements**. A matrix with m rows and n columns is called an $m \times n$ matrix.

Example of an **augmented** matrix:

$$\begin{cases} 3x_1 + 2x_2 - x_3 + x_4 = -1 \\ 2x_1 - x_3 + 2x_4 = 0 \\ 3x_1 + x_2 + 2x_3 + 5x_4 = 2 \end{cases} = \left[\begin{array}{cccc|c} 3 & 2 & -1 & 1 & -1 \\ 2 & 0 & -1 & 2 & 0 \\ 3 & 1 & 2 & 5 & 2 \end{array} \right]$$

Elementary Operations

Definition There are three types of **elementary row operations** that can be performed on a matrix:

- **Row swapping:** Interchanging two rows of the matrix.
- **Row scaling:** Multiplying all entries in a row by a non-zero constant.
- **Row addition:** Adding a multiple of one row to another row.

These operations are used in methods such as Gaussian elimination to solve systems of linear equations. Row operations do not produce equal matrices, but they do produce **equivalent** matrices, meaning they represent systems of equations with the same solution set. In order for 2 matrices to be equal, they must have the same dimensions and identical entries in corresponding positions.

Example Consider the matrices:

$$\left[\begin{array}{cc|c} 1 & 2 & 0 \\ 3 & 4 & 1 \end{array} \right] \mapsto \left[\begin{array}{cc|c} 2 & 4 & 0 \\ 3 & 4 & 1 \end{array} \right]$$

This represents the row operation of scaling the first row by 2 ($2R_1 \mapsto R_1$). These matrices are equivalent but not equal.

Example Solve:

$$\begin{aligned} \begin{cases} 2x + 3y - z = 3 \\ -x - y + z = 1 \\ x - 2y - 5z = 0 \end{cases} &= \left[\begin{array}{ccc|c} 2 & 3 & -1 & 3 \\ -1 & -1 & 1 & 1 \\ 1 & -2 & -5 & 0 \end{array} \right] \xrightarrow{R3 \mapsto R1} \left[\begin{array}{ccc|c} 1 & -2 & -5 & 0 \\ -1 & -1 & 1 & 1 \\ 2 & 3 & -1 & 3 \end{array} \right] \xrightarrow{R2+R1 \mapsto R2} \left[\begin{array}{ccc|c} 1 & -2 & -5 & 0 \\ 0 & -3 & -4 & 1 \\ 2 & 3 & -1 & 3 \end{array} \right] \\ &\xrightarrow{R3-2R1 \mapsto R3} \left[\begin{array}{ccc|c} 1 & -2 & -5 & 0 \\ 0 & -3 & -4 & 1 \\ 0 & 7 & 9 & 3 \end{array} \right] \xrightarrow{-\frac{1}{3}R2 \mapsto R2} \left[\begin{array}{ccc|c} 1 & -2 & -5 & 0 \\ 0 & 1 & \frac{4}{3} & -\frac{1}{3} \\ 0 & 7 & 9 & 3 \end{array} \right] \xrightarrow{R3-7R2 \mapsto R3} \left[\begin{array}{ccc|c} 1 & -2 & -5 & 0 \\ 0 & 1 & \frac{4}{3} & -\frac{1}{3} \\ 0 & 0 & \frac{-1}{3} & \frac{16}{3} \end{array} \right] \\ &\xrightarrow{-3R3 \mapsto R3} \left[\begin{array}{ccc|c} 1 & -2 & -5 & 0 \\ 0 & 1 & \frac{4}{3} & -\frac{1}{3} \\ 0 & 0 & 1 & -16 \end{array} \right] \xrightarrow{R2-\frac{4}{3}R3 \mapsto R2} \left[\begin{array}{ccc|c} 1 & -2 & -5 & 0 \\ 0 & 1 & 0 & 21 \\ 0 & 0 & 1 & -16 \end{array} \right] \xrightarrow{R1+5R3 \mapsto R1} \left[\begin{array}{ccc|c} 1 & -2 & 0 & -80 \\ 0 & 1 & 0 & 21 \\ 0 & 0 & 1 & -16 \end{array} \right] \\ &\xrightarrow{R1+2R2 \mapsto R1} \left[\begin{array}{ccc|c} 1 & 0 & 0 & -38 \\ 0 & 1 & 0 & 21 \\ 0 & 0 & 1 & -16 \end{array} \right] \implies (x, y, z) = (-38, 21, -16) \end{aligned}$$

1.2 Gaussian Elimination

Gaussian elimination is a method for solving systems of linear equations by transforming the system's augmented matrix into a simpler form using elementary row operations. The goal is to reach **row echelon form** or **reduced row echelon form**, from which the solutions can be easily obtained.

Row Echelon Form

Definition A matrix is in **row echelon form** if it satisfies the following conditions:

- All non-zero rows are above any rows of all zeros.
- The leading entry (the first non-zero number from the left, also called a **pivot**) of each non-zero row is to the right of the leading entry of the previous row.
- The leading entry in any non-zero row is 1 (this condition is sometimes omitted in basic row echelon form).

Example of a matrix in row echelon form:

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 1 & 3 & 4 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

It is often useful to further refine this to **reduced row echelon form**, which additionally requires that the leading 1 in each non-zero row is the only non-zero entry in its **column**.

Example of a matrix in reduced row echelon form:

$$\left[\begin{array}{ccc|c} 1 & -2 & 0 & 2 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Notes

- Every matrix can be reduced to row echelon form and reduced row echelon form using a sequence of elementary row operations.
- The reduced row echelon form of any matrix is unique.
- Row echelon form is not unique; different sequences of row operations can lead to different row echelon forms for the same matrix.

Zero Matrix A **zero matrix** is a matrix in which all entries are zero. It is often denoted by 0. The solution to the zero matrix is $\mathbb{R}^n$, where n is the number of variables.

Matrix Reduction Algorithm

1. Identify the leftmost non-zero column (the pivot column).
2. Select a non-zero entry in the pivot column as the pivot. If necessary, swap rows to move this entry to the top of the pivot column.
3. Scale the pivot row so that the pivot entry becomes 1.
4. Use row addition to create zeros in all positions below the pivot.
5. Cover the row containing the pivot and repeat the process for the submatrix that remains until all rows have been processed.
6. (Optional) To achieve reduced row echelon form, repeat the process to create zeros above each pivot as well.

Gaussian Elimination Algorithm

1. Augmented Matrix $[A|b]$ RREF with Row Operations
2. If a row of the form $[0 \ 0 \ \dots \ 0 \ | \ b]$ with b non-zero exists, then the system is inconsistent (no solutions).
3. Otherwise, assign free variables (if any) and solve for leading variables to find the solution set.
4. Write leading variables in terms of the parameters
5. If there are no nonleading variables, we have 1 solution. If there are nonleading variables, we have infinitely many solutions.

Example 1 Solve the system:

$$\begin{aligned}
 \begin{cases} 3x + y - 4z = 1 \\ x + 10z = 5 \\ 4x + y + 6z = 1 \end{cases} &= \left[\begin{array}{ccc|c} 3 & 1 & -4 & 1 \\ 1 & 0 & 10 & 5 \\ 4 & 1 & 6 & 1 \end{array} \right] \xrightarrow{R2 \leftrightarrow R1} \left[\begin{array}{ccc|c} 1 & 0 & 10 & 5 \\ 3 & 1 & -4 & 1 \\ 4 & 1 & 6 & 1 \end{array} \right] \xrightarrow{R2-3R1 \rightarrow R2} \left[\begin{array}{ccc|c} 1 & 0 & 10 & 5 \\ 0 & 1 & -34 & -14 \\ 4 & 1 & 6 & 1 \end{array} \right] \\
 &\xrightarrow{R3-4R1 \rightarrow R3} \left[\begin{array}{ccc|c} 1 & 0 & 10 & 5 \\ 0 & 1 & -34 & -14 \\ 0 & 1 & -34 & -19 \end{array} \right] \xrightarrow{R3-R2 \rightarrow R3} \left[\begin{array}{ccc|c} 1 & 0 & 10 & 5 \\ 0 & 1 & -34 & -14 \\ 0 & 0 & 0 & -5 \end{array} \right] \implies \text{no solutions}
 \end{aligned}$$

Example 2 Solve the system:

$$\begin{aligned}
 \begin{cases} x_1 - 2x_2 - x_3 + 3x_4 = 1 \\ 2x_1 - 4x_2 + x_3 = 5 \\ x_1 - 2x_2 + 2x_3 - 3x_4 = 4 \end{cases} &= \begin{bmatrix} 1 & -2 & -1 & 3 & | & 1 \\ 2 & -4 & 1 & 0 & | & 5 \\ 1 & -2 & 2 & -3 & | & 4 \end{bmatrix} \xrightarrow{R2-2R1 \rightarrow R2} \begin{bmatrix} 1 & -2 & -1 & 3 & | & 1 \\ 0 & 0 & 3 & -6 & | & 3 \\ 1 & -2 & 2 & -3 & | & 4 \end{bmatrix} \\
 &\xrightarrow{R3-R1 \rightarrow R3} \begin{bmatrix} 1 & -2 & -1 & 3 & | & 1 \\ 0 & 0 & 3 & -6 & | & 3 \\ 0 & 0 & 3 & -6 & | & 3 \end{bmatrix} \xrightarrow{R3-R2 \rightarrow R3} \begin{bmatrix} 1 & -2 & -1 & 3 & | & 1 \\ 0 & 0 & 3 & -6 & | & 3 \\ 0 & 0 & 0 & 0 & | & 0 \end{bmatrix} \\
 &\xrightarrow{\frac{1}{3}R2 \rightarrow R2} \begin{bmatrix} 1 & -2 & -1 & 3 & | & 1 \\ 0 & 0 & 1 & -2 & | & 1 \\ 0 & 0 & 0 & 0 & | & 0 \end{bmatrix} = \begin{cases} x_1 - 2x_2 - x_3 + 3x_4 = 1 \\ x_3 - 2x_4 = 1 \end{cases} \\
 &\Rightarrow \begin{cases} x_2 = s \\ x_4 = t \\ x_1 = 1 + 2s + t \\ x_3 = 1 + 2t \end{cases}
 \end{aligned}$$

Rank of a Matrix

Definition The **rank** of a matrix is the number of leading 1's (pivots) in its row echelon form or reduced row echelon form. It represents the dimension of the column space (or row space) of the matrix, which is the span of its columns (or rows). Rank is useful for finding the number of parameters in the solution set of a linear system.

Example 3 Consider the matrix from Example 2. Find the rank and number of parameters:

$$\begin{aligned}
 A = \begin{bmatrix} 1 & -2 & -1 & 3 & | & 1 \\ 2 & -4 & 1 & 0 & | & 5 \\ 1 & -2 & 2 & -3 & | & 4 \end{bmatrix} &\xrightarrow{\text{RREF}(A)} \begin{bmatrix} 1 & -2 & 0 & 1 & | & 0 \\ 0 & 0 & 1 & -2 & | & 1 \\ 0 & 0 & 0 & 0 & | & 0 \end{bmatrix} \Rightarrow \text{rank}(A) = 2 \\
 n = 4 \text{ (number of variables)} &\Rightarrow \text{number of parameters} = n - \text{rank}(A) = 4 - 2 = 2
 \end{aligned}$$

Example 4 Find the rank of the following matrices:

$$\begin{aligned}
 A = \begin{bmatrix} 1 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} &\Rightarrow \text{rank}(A) = 2 \\
 B = \begin{bmatrix} 0 & 0 \\ 0 & 3 \end{bmatrix} &\xrightarrow{R1 \leftrightarrow R2} \begin{bmatrix} 0 & 3 \\ 0 & 0 \end{bmatrix} \xrightarrow{\frac{1}{3}R1 \rightarrow R1} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \Rightarrow \text{rank}(B) = 1 \\
 C = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} &\Rightarrow \text{rank}(C) = 0
 \end{aligned}$$

1.3 Homogeneous Systems

Definition A system is called **homogeneous** if all the constant terms are zero. In other words, a system of equations is homogeneous if it can be written in the form

$$\begin{cases} x - 5y = 0 \\ 2x + 3y + z = 0 \end{cases}$$

The solution set of a homogeneous system always contains the **trivial solution**, which is the solution where all variables are equal to zero. For example, in the system above, the trivial solution is $x = 0, y = 0, z = 0$. Any other solution is called a **nontrivial solution**.

Example Find any non-trivial solution

$$\begin{aligned} & \begin{bmatrix} 1 & -1 & 2 & -1 & | & 0 \\ 2 & 2 & 0 & 1 & | & 0 \\ 3 & 1 & 2 & -1 & | & 0 \end{bmatrix} \xrightarrow{R_2 \mapsto R_2 - 2R_1} \begin{bmatrix} 1 & -1 & 2 & -1 & | & 0 \\ 0 & 4 & -4 & 3 & | & 0 \\ 3 & 1 & 2 & -1 & | & 0 \end{bmatrix} \\ & \xrightarrow{R_3 \mapsto R_3 - 3R_1} \begin{bmatrix} 1 & -1 & 2 & -1 & | & 0 \\ 0 & 4 & -4 & 3 & | & 0 \\ 0 & 4 & -4 & 2 & | & 0 \end{bmatrix} \xrightarrow{R_3 \mapsto R_3 - R_2} \begin{bmatrix} 1 & -1 & 2 & -1 & | & 0 \\ 0 & 4 & -4 & 3 & | & 0 \\ 0 & 0 & 0 & -1 & | & 0 \end{bmatrix} \\ & \xrightarrow{R_1 \mapsto R_1 - R_3} \begin{bmatrix} 1 & -1 & 2 & 0 & | & 0 \\ 0 & 4 & -4 & 3 & | & 0 \\ 0 & 0 & 0 & -1 & | & 0 \end{bmatrix} \xrightarrow{R_2 \mapsto R_2 + 3R_3} \begin{bmatrix} 1 & -1 & 2 & 0 & | & 0 \\ 0 & 4 & -4 & 0 & | & 0 \\ 0 & 0 & 0 & -1 & | & 0 \end{bmatrix} \\ & \xrightarrow{R_3 \mapsto -R_3} \begin{bmatrix} 1 & -1 & 2 & 0 & | & 0 \\ 0 & 4 & -4 & 0 & | & 0 \\ 0 & 0 & 0 & 1 & | & 0 \end{bmatrix} \xrightarrow{R_2 \mapsto \frac{1}{4}R_2} \begin{bmatrix} 1 & -1 & 2 & 0 & | & 0 \\ 0 & 1 & -1 & 0 & | & 0 \\ 0 & 0 & 0 & 1 & | & 0 \end{bmatrix} \\ & \xrightarrow{R_1 \mapsto R_1 + R_2} \begin{bmatrix} 1 & 0 & 1 & 0 & | & 0 \\ 0 & 1 & -1 & 0 & | & 0 \\ 0 & 0 & 0 & 1 & | & 0 \end{bmatrix} \Rightarrow \begin{cases} x_1 + x_3 = 0 \\ x_2 - x_3 = 0 \\ x_4 = 0 \end{cases} = \begin{cases} x_1 = -x_3 \\ x_2 = x_3 \\ x_4 = 0 \end{cases} \\ & x_3 = S \Rightarrow \begin{cases} x_1 = -S \\ x_2 = S \\ x_3 = S \\ x_4 = 0 \end{cases} \quad S \in \mathbb{R} \quad S = 1 \Rightarrow \begin{cases} x_1 = -1 \\ x_2 = 1 \\ x_3 = 1 \\ x_4 = 0 \end{cases} \end{aligned}$$

Theorem If a homogeneous system of linear equations has ore variables than equations, then it has infinitely many non-trivial solutions. For some matrix A with m rows and n parameters $\text{rank}(A) \leq m < n$ implies that the solutions involved $n - \text{rank}(A) \geq 1$ parameters, so $x_i = S$ and we can choose any value for S other than zero to get a non-trivial solution.

Note If a homogeneous system has a nontrivial solution, it need not have more variables than equations.

Note If a homogeneous system has a non-trivial solution, then it has infinitely many non-trivial solutions. If $x = S$ is a non-trivial solution, then so is $x = kS$ for any scalar $k \neq 0$.

Vectors

Definition A **vector** in $\mathbb{R}^n$ is an ordered n -tuple of real numbers. We can represent a vector as a column matrix:

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = (x_1, x_2, \dots, x_n) = \langle x_1, x_2, \dots, x_n \rangle$$

where $x_1, x_2, \dots, x_n$ are the components of the vector $\vec{x}$. In this class, vectors should always be represented as column matrices unless otherwise specified.

Definition A vector of length n is an ordered n -tuple, and the set of all vectors of length n is denoted by $\mathbb{R}^n$. In other words, $\vec{x} \in \mathbb{R}^n$ if $\vec{x}$ has n components.

Vector Addition Vector addition can only be performed on vectors of the same length. If $\vec{x}, \vec{y} \in \mathbb{R}^n$, then their sum is given by

$$\vec{x} + \vec{y} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{bmatrix}$$

Scalar Multiplication If $\vec{x} \in \mathbb{R}^n$ and c is a scalar, then the scalar multiplication of c and $\vec{x}$ is given by

$$c\vec{x} = c \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} cx_1 \\ cx_2 \\ \vdots \\ cx_n \end{bmatrix}$$

Linear Combination A sum of scalar multiples of several vectors or columns is called a **linear combination** of the vectors or columns. For example, if $\vec{u}, \vec{v}, \vec{w} \in \mathbb{R}^n$ and a, b, c are scalars, then

$$a\vec{u} + b\vec{v} + c\vec{w}$$

is a linear combination of the vectors $\vec{u}, \vec{v}, \vec{w}$.

Example Let $\vec{x} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, $\vec{y} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$, $\vec{z} = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$, $\vec{v} = \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix}$, $\vec{w} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.

a. Find $2\vec{x} - \vec{y} + 3\vec{z}$, b. Determine if $\vec{v}$ is a linear combination of $\vec{x}, \vec{y}, \vec{z}$, and c. Determine if $\vec{w}$ is a linear combination of $\vec{x}, \vec{y}, \vec{z}$.

$$a. \quad 2\vec{x} - \vec{y} + 3\vec{z} = 2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 9 \\ 5 \\ 3 \end{bmatrix}$$

$$b. \quad \exists r, s, t \in \mathbb{R} \ni r\vec{x} + s\vec{y} + t\vec{z} = \vec{v} \implies \begin{bmatrix} r \\ 0 \\ r \end{bmatrix} + \begin{bmatrix} 2s \\ s \\ 0 \end{bmatrix} + \begin{bmatrix} 3t \\ t \\ t \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix}$$

$$\implies \begin{cases} r + 2s + 3t = 0 \\ s + t = -1 \\ r + t = 2 \end{cases} \implies \left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & 1 & -1 \\ 1 & 0 & 1 & 2 \end{array} \right] \xrightarrow{RREF} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\implies \begin{cases} r + t = 2 \\ s + t = -1 \end{cases} \implies \begin{cases} r = 2 - q \\ s = -1 - q \\ t = q \end{cases} \quad t \in \mathbb{R} \quad q = 0 \implies \begin{cases} r = 2 \\ s = -1 \\ t = 0 \end{cases} \implies \vec{v} = 2\vec{x} - \vec{y}$$

$$c. \quad \exists r, s, t \in \mathbb{R} \ni r\vec{x} + s\vec{y} + t\vec{z} = \vec{w} \implies \left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{array} \right] \xrightarrow{RREF} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & -1 \end{array} \right]$$

$\implies$ No solution, so $\vec{w}$ is not a linear combination of $\vec{x}, \vec{y}, \vec{z}$

Note To find if $\vec{x}$ is a linear combination of $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_n$, we can form the augmented matrix

$$[\vec{a}_1 \ \vec{a}_2 \ \dots \ \vec{a}_n \mid \vec{x}]$$

and row reduce to see if there is a solution. Linear combinations are a good way to write the general solution to a homogeneous system.

Theorem Any linear combination of solutions to a homogeneous system is also a solution to the system.

Example Let $\vec{p} = \begin{bmatrix} p_1 \\ p_2 \end{bmatrix}$ and $\vec{q} = \begin{bmatrix} q_1 \\ q_2 \end{bmatrix}$ be solutions to the homogeneous system $a_1x_1 + a_2x_2 = 0$, hence:

$$\implies \exists s, t \in \mathbb{R} \ni \begin{cases} a_1p_1 + a_2p_2 = 0 \\ a_1q_1 + a_2q_2 = 0 \end{cases} = s\vec{p} + t\vec{q} = \begin{bmatrix} sp_1 + tq_1 \\ sp_2 + tq_2 \end{bmatrix} \text{ is a solution}$$

$$a_1(sp_1 + tq_1) + a_2(sp_2 + tq_2) = 0 \implies s \cdot 0 + t \cdot 0 = 0$$

Example Solve the homogeneous system w/ coefficient matrix $A = \begin{bmatrix} 1 & -2 & 3 & -2 \\ -3 & 6 & 1 & 0 \\ -2 & 4 & 4 & -2 \end{bmatrix}$

$$RREF(A) = \left[\begin{array}{cccc|c} 1 & -2 & 0 & -\frac{1}{5} & 0 \\ 0 & 0 & 1 & -\frac{3}{5} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \quad \text{Rank}(A) = 2 \implies 4 - 2 = 2 \text{ parameters}$$

$$\implies \begin{cases} x_1 = 2s + \frac{1}{5}t \\ x_2 = s \\ x_3 = \frac{3}{5}t \\ x_4 = t \end{cases} = \begin{bmatrix} 2s + \frac{1}{5}t \\ s \\ \frac{3}{5}t \\ t \end{bmatrix} = s \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} \frac{1}{5} \\ 0 \\ \frac{3}{5} \\ 1 \end{bmatrix} \text{ is the general solution}$$

so our basic solutions are $\begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} \frac{1}{5} \\ 0 \\ \frac{3}{5} \\ 1 \end{bmatrix}$

Note Every solution to a homogeneous system can be written as a linear combination of the basic solutions.

2. Matrices and Matrix Operations

2.1 Matrices and Matrix Operations

Definition A **matrix** is a rectangular array of numbers arranged in rows and columns. An $m \times n$ matrix has m rows and n columns. For example, the following is a 2×3 matrix:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

Matrix Access The entry in the i -th row and j -th column of a matrix A is denoted by a_{ij} . For example, in the matrix A above, $a_{12} = 2$ and $a_{21} = 4$.

$$A_{m \times n} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

Matrix Operations

Matrix Equality Two matrices A and B are equal if they have the same dimensions and their corresponding entries are equal. That is, $A = B$ if and only if $a_{ij} = b_{ij}$ for all i and j .

Matrix Addition Two matrices can be added if they have the same dimensions, otherwise it is undefined. If A and B are both $m \times n$ matrices, then their sum is given by

$$A + B = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \dots & a_{1n} + b_{1n} \\ a_{21} + b_{21} & a_{22} + b_{22} & \dots & a_{2n} + b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} + b_{m1} & a_{m2} + b_{m2} & \dots & a_{mn} + b_{mn} \end{bmatrix}$$

Scalar Multiplication If A is an $m \times n$ matrix and c is a scalar, then the scalar multiplication of c and A is given by

$$cA = \begin{bmatrix} ca_{11} & ca_{12} & \dots & ca_{1n} \\ ca_{21} & ca_{22} & \dots & ca_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ ca_{m1} & ca_{m2} & \dots & ca_{mn} \end{bmatrix}$$

Properties of Matrix Addition and Scalar Multiplication Let A , B , and C be matrices of the same dimensions, and let c and d be scalars. Then the following properties hold:

- Commutative Property: $A + B = B + A$
- Associative Property: $(A + B) + C = A + (B + C)$

- Distributive Property: $c(A + B) = cA + cB$
- Scalar Distributive Property: $(c + d)A = cA + dA$
- Associative Property of Scalar Multiplication: $c(dA) = (cd)A$
- Identity Property: $A + 0 = A$, where 0 is the zero matrix of the same dimensions as A

Matrix Transpose The transpose of a matrix A , denoted by A^T , is obtained by interchanging its rows and columns. If A is an $m \times n$ matrix, then A^T is an $n \times m$ matrix given by

$$A^T = \begin{bmatrix} a_{11} & a_{21} & \dots & a_{m1} \\ a_{12} & a_{22} & \dots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \dots & a_{mn} \end{bmatrix}$$

Transposition Identities Let A, B be $m \times n$ matrices and $k \in \mathbb{R}$

- A^T is $n \times m$
- $(A^T)^T = A$
- $(kA)^T = k(A^T)$
- $(A + B)^T = A^T + B^T$

Example Solve for A

$$(2A^T + 3 \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix})^T$$

Symmetric Matrix A matrix A is called symmetric if $A = A^T$. If A, B are $n \times n$ symmetric matrices, then $A + B$ is also symmetric. If c is a scalar, then cA is also symmetric. If A and B are both symmetric, then $kA + pB$ is also symmetric for $k, p \in \mathbb{R}$.

2.2 Matrix-Vector Multiplication

Vector-Matrix Multiplication

Definition Let A be an $m \times n$ matrix written $A = [a_1 \ a_2 \ \dots \ a_n]$ where a_i is the i -th column of A . Let $\vec{x}$ be an $n \times 1$ column vector written as $\vec{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$. Then the matrix-vector product $A\vec{x}$ is defined as

$$\begin{aligned} A\vec{x} &= [a_1 \ a_2 \ \dots \ a_n] \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \\ &= x_1 a_1 + x_2 a_2 + \dots + x_n a_n \\ &= \sum_{i=1}^n x_i a_i \end{aligned}$$

Notes

- If A is an $m \times n$ matrix, then $A\vec{x}$ is defined only for $\vec{x} \in \mathbb{R}^n$.
- The inner sizes of the vector and matrix must match for the product to be defined.
- $A\vec{x}$ is a linear combination of the columns of A with coefficients x_i in order.

Example Let $B = \begin{bmatrix} 1 & 0 & 3 \\ 2 & -1 & 1 \end{bmatrix}$ and $\vec{q} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$. Find $B\vec{q}$.

$$B\vec{q} = 1 \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix} + 2 \cdot \begin{bmatrix} 0 \\ -1 \end{bmatrix} + 3 \cdot \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ -2 \end{bmatrix} + \begin{bmatrix} 9 \\ 3 \end{bmatrix} = \begin{bmatrix} 10 \\ 3 \end{bmatrix}$$

Example Let $A = \begin{bmatrix} 0 & -1 & 2 \\ 5 & 3 & -4 \\ 1 & 0 & 1 \end{bmatrix}$ and $\vec{p} = \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix}$. Find $A\vec{p}$.

$$A\vec{p} = 1 \cdot \begin{bmatrix} 0 \\ 5 \\ 1 \end{bmatrix} + 0 \cdot \begin{bmatrix} -1 \\ 3 \\ 0 \end{bmatrix} + (-3) \cdot \begin{bmatrix} 2 \\ -4 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} -6 \\ 12 \\ -3 \end{bmatrix} = \begin{bmatrix} -6 \\ 17 \\ -2 \end{bmatrix}$$

Theorem A system of linear equations $\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m \end{cases}$ can be written in

matrix form as $A\vec{x} = \vec{b}$ where

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}, \quad \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \vec{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

A is called the coefficient matrix, $\vec{x}$ is called the variable vector, and $\vec{b}$ is called the constant vector. The system of equations has a solution if and only if there exists a vector $\vec{x}$ such that $A\vec{x} = \vec{b}$.

- Every system of linear equations can be written as $A\vec{x} = \vec{b}$
- $A\vec{x} = \vec{b}$ is consistent if and only if $\vec{b}$ is a linear combination of columns of A .

Identity Matrix

Definition The $n \times n$ identity matrix, denoted by I_n , is the square matrix with 1's on the main diagonal and 0's elsewhere. That is,

$$I_n = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix}$$

Properties of the Identity Matrix

- $\text{rank}(I_n) = n$
- $I_n\vec{x} = \vec{x}$ for all $\vec{x} \in \mathbb{R}^n$

More Identities Let A and B be $n \times n$ matrices and $\vec{x}, \vec{y} \in \mathbb{R}^n$

- $A(\vec{x} + \vec{y}) = A\vec{x} + A\vec{y}$
- $A(k\vec{x}) = k(A\vec{x})$ for $k \in \mathbb{R}$
- $(A + B)\vec{x} = A\vec{x} + B\vec{x}$

Theorem Let $A\vec{x} = \vec{b}$ be a system of linear equations and $\text{rank}(A) = r$. Then the rank of the augmented matrix $[A|\vec{b}]$ is either r or $r + 1$. The system is consistent if and only if $\text{rank}(A) = \text{rank}([A|\vec{b}])$.

Example Let $A = \begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ and $\vec{b} = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \end{bmatrix}$. Determine if the system $A\vec{x} = \vec{b}$ is consistent.

$\text{rank}(A) = 2$ and $\text{rank}([A|\vec{b}]) = 3 \implies$ the system is inconsistent

Dot Product Let $\vec{a} = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}$ and $\vec{b} = \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}$ be two vectors in $\mathbb{R}^n$. The dot product of $\vec{a}$ and $\vec{b}$, denoted by $\vec{a} \cdot \vec{b}$, is defined as

$$\vec{a} \cdot \vec{b} = a_1b_1 + \dots + a_nb_n = \sum_{i=1}^n a_ib_i$$

The result of the dot product is a scalar.

Definition Let A be an $m \times n$ matrix and $\vec{x} \in \mathbb{R}^n$, then the i th entry of the vector $A\vec{x}$ is the dot product of the i th row of A with $\vec{x}$

Example Let $A = \begin{bmatrix} 1 & -1 & 3 & -2 \\ 2 & 3 & 5 & 7 \\ 3 & 0 & 4 & 0 \\ 4 & 5 & 2 & 0 \end{bmatrix}$ and $\vec{x} = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 5 \end{bmatrix}$. Find the second entry of $A\vec{x}$ without computing the entire product.

$$P_2 = [2 \ 3 \ 5 \ 7] \cdot \begin{bmatrix} 0 \\ 1 \\ -1 \\ 5 \end{bmatrix} = 2(0) + 3(1) + 5(-1) + 7(5) = 3 - 5 + 35 = 33$$

Example Let $C = \begin{bmatrix} 2 & -1 & 3 & 5 \\ 0 & 2 & -3 & 1 \\ -3 & 4 & 1 & 2 \end{bmatrix}$ and $\vec{q} = \begin{bmatrix} 2 \\ 1 \\ 0 \\ -1 \end{bmatrix}$. Find $C\vec{q}$.

$$\begin{aligned} C\vec{q} &= 2 \cdot \begin{bmatrix} 2 \\ 0 \\ -3 \end{bmatrix} + 1 \cdot \begin{bmatrix} -1 \\ 2 \\ 4 \end{bmatrix} + 0 \cdot \begin{bmatrix} 3 \\ -3 \\ 1 \end{bmatrix} + (-1) \cdot \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} 4 \\ 0 \\ -6 \end{bmatrix} + \begin{bmatrix} -1 \\ 2 \\ 4 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} -5 \\ -1 \\ -2 \end{bmatrix} \\ &= \begin{bmatrix} -2 \\ 1 \\ -4 \end{bmatrix} \end{aligned}$$

Standard Basis Vectors The standard basis vectors in $\mathbb{R}^n$ are the vectors $e_1, e_2, \dots, e_n$ where e_i has a 1 in the i -th position and 0's elsewhere. That is,

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \quad e_i = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

We can use these basis vectors to isolate the columns of a matrix. For example, if $A = [a_1 \ a_2 \ \dots \ a_n]$ is an $m \times n$ matrix, then

$$\begin{aligned} Ae_1 &= a_1 \\ Ae_2 &= a_2 \\ &\vdots \\ Ae_n &= a_n \end{aligned}$$

Theorem $A, B \in \mathbb{R}^{m \times n}, \vec{x} \in \mathbb{R}^n \implies A = B \implies \forall i < n \in \mathbb{N}, Ae_i = Be_i$

Note $A\vec{x} = B\vec{x}$ does not imply $A = B$ for all $\vec{x} \in \mathbb{R}^n$. For example, if A and B are both $m \times n$ matrices with all entries equal to 0, then $A\vec{x} = B\vec{x}$ for all $\vec{x} \in \mathbb{R}^n$, but A and B are not necessarily the same matrix.

Theorem $A\vec{x} = B\vec{x} \implies \forall \vec{x} \in \mathbb{R}^n, A\vec{x} = B\vec{x}$

Transformations

Definition Functions $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ are called transformations. The set $\mathbb{R}^n$ is called the domain of T and the set $\mathbb{R}^m$ is called the codomain of T . The image of a vector $\vec{x} \in \mathbb{R}^n$ under the transformation T is denoted by $T(\vec{x})$.

Matrix Transformations A matrix transformation is denoted by $T_A(\vec{x}) = A\vec{x}$ for some matrix $A_{m \times n}$. If $A \in \mathbb{R}^{m \times n}$, then $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$

Example $\forall \vec{x} \in \mathbb{R}^2, T_A(\vec{x}) = A\vec{x}$ and let $A = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix}$ and $\vec{u} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$. Find $T_A(\vec{u})$.

$$T_A(\vec{u}) = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 1(2) + (-3)(-1) \\ 3(2) + 5(-1) \\ -1(2) + 7(-1) \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \\ -9 \end{bmatrix}$$

Example Let $\vec{b} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}$ and $\vec{c} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$. Find $\vec{x} \in \mathbb{R}^2 \ni T_A(\vec{x}) = \vec{b}$ and $\vec{y} \in \mathbb{R}^2 \ni T_A(\vec{y}) = \vec{c}$.

$$\begin{aligned}
 A\vec{x} = \vec{b} &\implies \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix} \implies \left[\begin{array}{cc|c} 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & -5 \end{array} \right] \xrightarrow{RREF} \left[\begin{array}{cc|c} 1 & 0 & \frac{3}{2} \\ 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 0 \end{array} \right] \\
 &\implies \vec{x} = \begin{bmatrix} \frac{3}{2} \\ -\frac{1}{2} \end{bmatrix}
 \end{aligned}$$

$$A\vec{y} = \vec{c} \implies \left[\begin{array}{cc|c} 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & 5 \end{array} \right] \xrightarrow{RREF} \left[\begin{array}{cc|c} 1 & -3 & 3 \\ 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 10 \end{array} \right] \implies \text{the system is inconsistent}$$

2.3 Matrix-Matrix Multiplication

Definition $\forall A \in \mathbb{R}^{m \times n}, B \in \mathbb{R}^{n \times p} \implies AB \in \mathbb{R}^{m \times p} = \begin{bmatrix} A\vec{b}_1 & A\vec{b}_2 & \dots & A\vec{b}_p \end{bmatrix}$ where $\vec{b}_i$ is the i -th column of B .